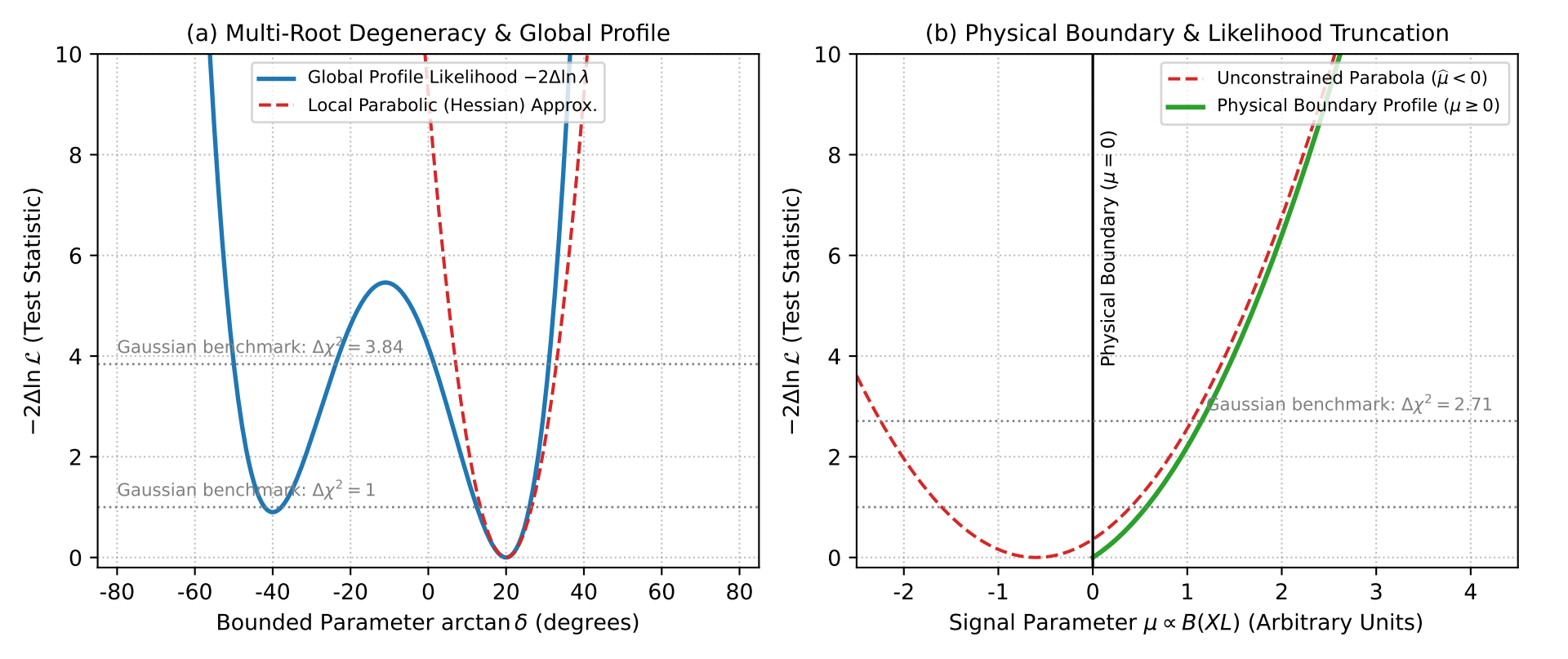
(a) Multi-Root Degeneracy & Global Profile
Global Profile Likelihood $-2\Delta\ln\lambda$
Local Parabolic (Hessian) Approx.
Gaussian benchmark: $\Delta\chi^2 = 3.84$
Gaussian benchmark: $\Delta\chi^2 = 1$
$-2\Delta\ln\mathcal{L}$ (Test Statistic)
Bounded Parameter $\arctan\delta$ (degrees)
(b) Physical Boundary & Likelihood Truncation
Unconstrained Parabola ($\hat{\mu} < 0$)
Physical Boundary Profile ($\mu \geq 0$)
Physical Boundary ($\mu = 0$)
Gaussian benchmark: $\Delta\chi^2 = 2.71$
$-2\Delta\ln\mathcal{L}$ (Test Statistic)
Signal Parameter $\mu \propto B(XL)$ (Arbitrary Units)